Physical Chemistry (I)

Lecture 12

Helmholtz and Gibbs Energies

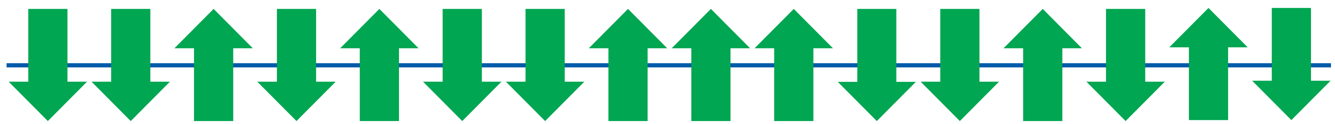
In the Last Class

- Statistical meaning of entropy
  - Spin systems: negative temperature
  - Shannon entropy
- Helmholtz and Gibbs energies
  - Definitions
  - Physical meanings

Spins

Electrons and protons have spins: up (↑) or down (↓)

Consider a solid system consisting of N spins:



- Energy splitting in a magnetic field

$$E(\uparrow) = +\frac{\epsilon}{2}, \qquad E(\downarrow) = -\frac{\epsilon}{2}$$

Macrostate

- Macrostate: number of "up" spins ($N_\uparrow$)
- Internal energy

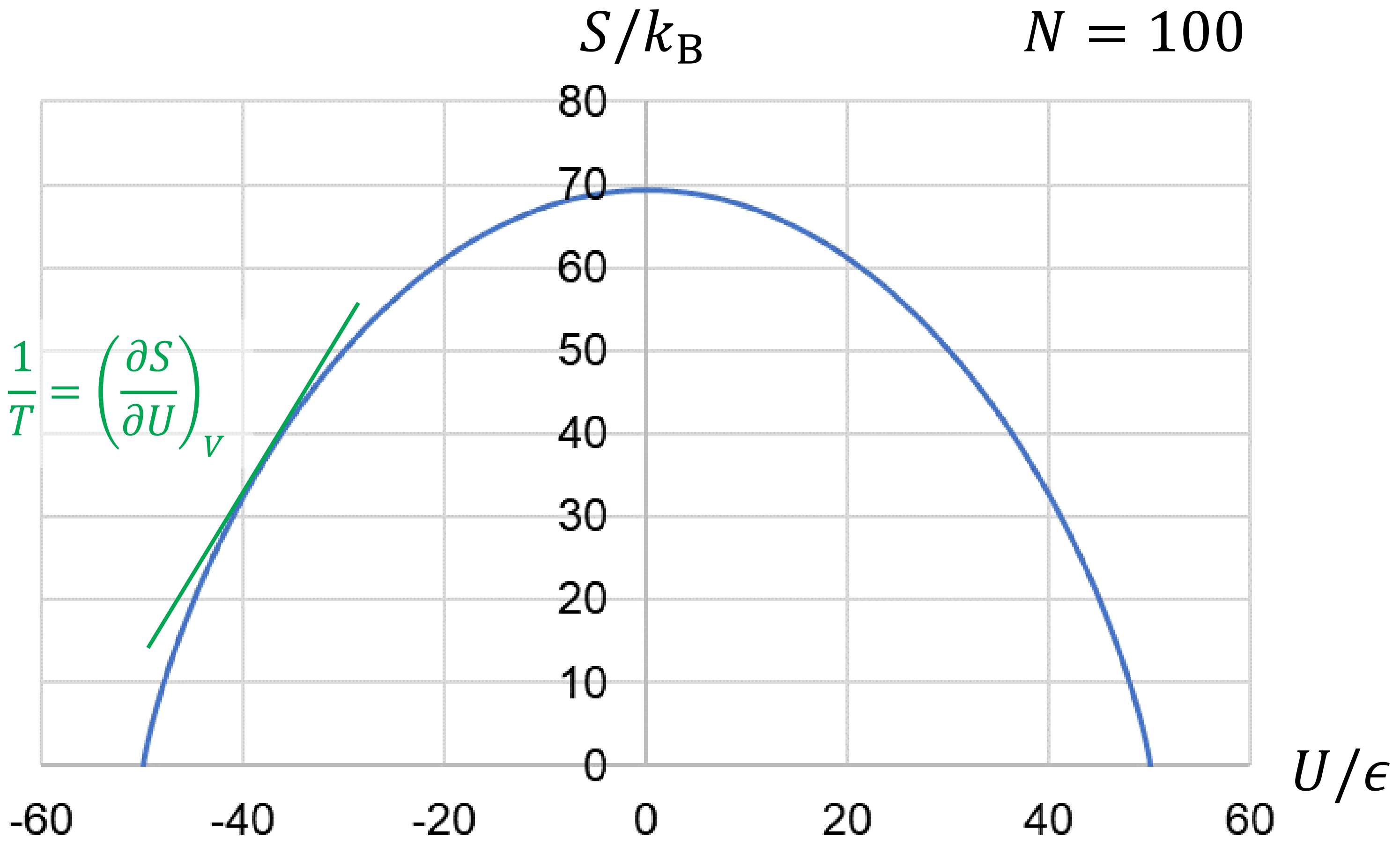
$$U = \epsilon N_\uparrow - \frac{\epsilon}{2} N$$

- Entropy

$$S = k_\mathrm{B}[N \ln N - N_\uparrow \ln N_\uparrow - (N - N_\uparrow) \ln(N - N_\uparrow)]$$

Last class

Entropy and Temperature



Shannon Entropy

Consider a system with the following properties:

- t microstates
- each microstate i with probability p_i
- N independent replicates ($N \to \infty$)

From the entropy of the ensemble, we can obtain the entropy of a system as

$$S = -k_\mathrm{B} \sum_i p_i \ln p_i$$

Last class

Helmholtz and Gibbs Energies

Helmholtz energy $A = U - TS$

Gibbs energy $G = H - TS$

Criteria of spontaneity:

$$\mathrm{d}A_{T,V} \leq 0$$

(At constant T and V, $\mathrm{d}A \propto -\mathrm{d}S_{\text{total}}$)

$$\mathrm{d}G_{T,p} \leq 0$$

(At constant T and p, $\mathrm{d}G \propto -\mathrm{d}S_{\text{total}}$)

So, What Do They Mean?

- Helmholtz energy = maximum work
- Gibbs energy = maximum non-expansion work

Thermodynamic Potentials

$+pV$

$-TS$

U	$H = U + pV$
$A = U - TS$	$G = U + pV - TS$

Differential Forms

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$ (fundamental equation)

- Enthalpy $H = U + pV$

$$\mathrm{d}H = \mathrm{d}U + p\mathrm{d}V + V\mathrm{d}p = (T\mathrm{d}S - \cancel{p\mathrm{d}V}) + \cancel{p\mathrm{d}V} + V\mathrm{d}p$$

$$\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$$

- Helmholtz energy $A = U - TS$

$$\mathrm{d}A = \mathrm{d}U - T\mathrm{d}S - S\mathrm{d}T = (\cancel{T\mathrm{d}S} - p\mathrm{d}V) - \cancel{T\mathrm{d}S} - S\mathrm{d}T$$

$$\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V$$

Differential Forms

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$ (fundamental equation)

- Gibbs energy $G = U + pV - TS$

$$\mathrm{d}G = \mathrm{d}U + p\mathrm{d}V + V\mathrm{d}p - T\mathrm{d}S - S\mathrm{d}T$$

$$\mathrm{d}G = (\cancel{T\mathrm{d}S} - \cancel{p\mathrm{d}V}) + \cancel{p\mathrm{d}V} + V\mathrm{d}p - \cancel{T\mathrm{d}S} - S\mathrm{d}T$$

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p$$

Natural Independent Variables

- Internal energy $U = U(S, V)$

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$

- Enthalpy $H = U + pV = H(S, p)$

$$\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$$

- Helmholtz energy $A = U - TS = A(T, V)$

$$\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V$$

- Gibbs energy $G = U + pV - TS = G(T, p)$

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p$$

Not in Atkins

Exact Differentials

The thermodynamic potentials are state functions

For $H = H(S, p)$,

$$\mathrm{d}H = \left(\frac{\partial H}{\partial S}\right)_p \mathrm{d}S + \left(\frac{\partial H}{\partial p}\right)_S \mathrm{d}p$$

Comparing this to $\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$,

$$\left(\frac{\partial H}{\partial S}\right)_p = T, \qquad \left(\frac{\partial H}{\partial p}\right)_S = V$$

Not in Atkins

Exact Differentials

Similarly, using $A = A(T, V)$, one can show

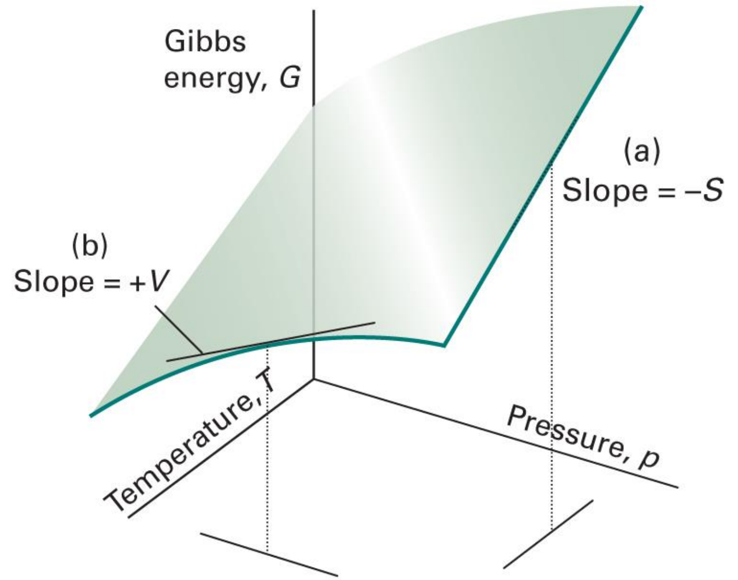
$$\left(\frac{\partial A}{\partial T}\right)_V = -S, \qquad \left(\frac{\partial A}{\partial V}\right)_T = -p$$

Using $G = G(T, p)$, one can show

$$\left(\frac{\partial G}{\partial T}\right)_p = -S, \qquad \left(\frac{\partial G}{\partial p}\right)_T = V$$

Example: Gibbs Energy

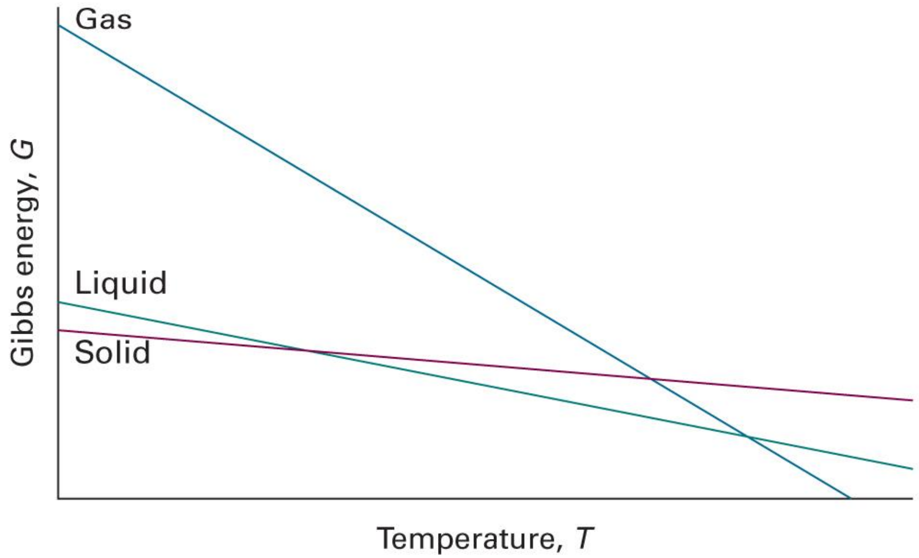
$$\left(\frac{\partial G}{\partial T}\right)_p = -S, \qquad \left(\frac{\partial G}{\partial p}\right)_T = V$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 3E.1)

Physical Interpretation

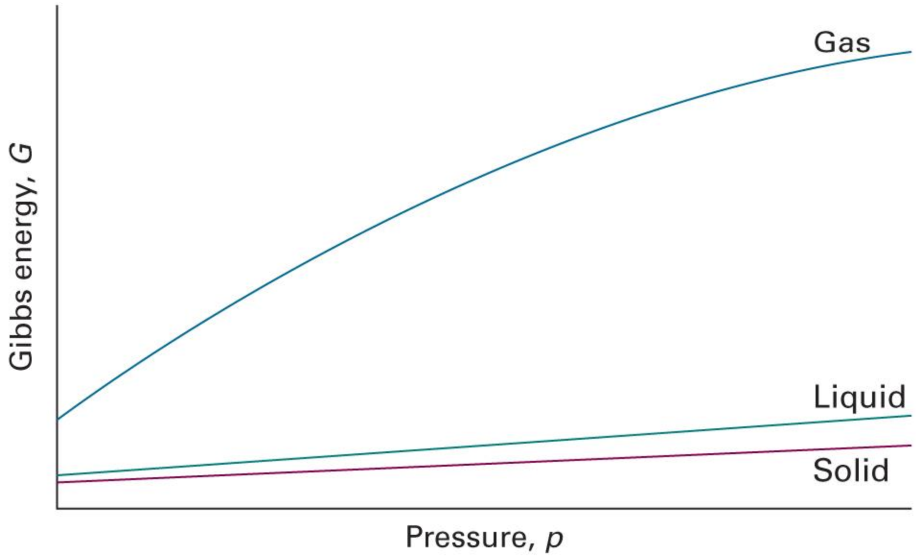
$$\left(\frac{\partial G}{\partial T}\right)_p = -S < 0$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 3E.2)

Physical Interpretation

$$\left(\frac{\partial G}{\partial p}\right)_T = V > 0$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 3E.3)

$\boldsymbol{G}$ and $\boldsymbol{T}$

$$\left(\frac{\partial G}{\partial T}\right)_p = -S$$

Since $G = H - TS$,

$$\left(\frac{\partial G}{\partial T}\right)_p = \frac{G - H}{T}$$

Gibbs-Helmholtz Equation

$$\left(\frac{\partial(G/T)}{\partial T}\right)_p = \left[\frac{\partial}{\partial T}\left(G \times \frac{1}{T}\right)\right]_p = \frac{1}{T}\left(\frac{\partial G}{\partial T}\right)_p + G\left[\frac{\partial}{\partial T}\left(\frac{1}{T}\right)\right]_p$$

$$\left(\frac{\partial(G/T)}{\partial T}\right)_p = \frac{1}{T}\left(\frac{\partial G}{\partial T}\right)_p - \frac{G}{T^2}$$

Since $(\partial G/\partial T)_p = (G - H)/T$,

$$\left(\frac{\partial(G/T)}{\partial T}\right)_p = \frac{G - H}{T^2} - \frac{G}{T^2} = -\frac{H}{T^2}$$

Application of G-H Equation

$$\left(\frac{\partial(G/T)}{\partial T}\right)_p = -\frac{H}{T^2}$$

At constant T,

$$\left(\frac{\partial(\Delta G/T)}{\partial T}\right)_p = -\frac{\Delta H}{T^2}$$

Review

Property 1 of Exact Differential

If f is a function of x and y, the differential $\mathrm{d}f$ is

$$\mathrm{d}f = \left(\frac{\partial f}{\partial x}\right)_y \mathrm{d}x + \left(\frac{\partial f}{\partial y}\right)_x \mathrm{d}y$$

when $\mathrm{d}f$ is an exact differential.

Then, we have

$$\left(\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)_y\right)_x = \left(\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)_x\right)_y$$

(Schwartz's theorem)

Maxwell Relations

For an exact differential $\mathrm{d}f = g\mathrm{d}x + h\mathrm{d}y$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Internal energy $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$

- Enthalpy $\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

Maxwell Relations

For an exact differential $\mathrm{d}f = g\mathrm{d}x + h\mathrm{d}y$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Helmholtz energy $\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

- Gibbs energy $\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p$

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$$

Example 3E.1

Use the Maxwell relations to show that the entropy of a perfect gas is linearly dependent on $\ln V$, that is, $S = a + b \ln V$.

From the Maxwell relations,

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

For a perfect gas,

$$\left(\frac{\partial p}{\partial T}\right)_V = \left[\frac{\partial}{\partial T}\left(\frac{nRT}{V}\right)\right]_V = \frac{nR}{V}$$

Hence,

$$\left(\frac{\partial S}{\partial V}\right)_T = \frac{nR}{V}$$

Example 3E.1

$$\left(\frac{\partial S}{\partial V}\right)_T = \frac{nR}{V}$$

At constant temperature,

$$\mathrm{d}S = \frac{nR}{V}\mathrm{d}V$$

$$\int \mathrm{d}S = \int \frac{nR}{V}\mathrm{d}V$$

Therefore,

$$S = nR \ln V + C$$

Review

Property 2 of Exact Differential

If f is a function of x and y, the differential $\mathrm{d}f$ is

$$\mathrm{d}f = \left(\frac{\partial f}{\partial x}\right)_y \mathrm{d}x + \left(\frac{\partial f}{\partial y}\right)_x \mathrm{d}y$$

when $\mathrm{d}f$ is an exact differential.

Differentiating with respect to x at constant z gives

$$\left(\frac{\partial f}{\partial x}\right)_z = \left(\frac{\partial f}{\partial x}\right)_y + \left(\frac{\partial f}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_z$$

Internal Pressure

Internal pressure $\pi_T = (\partial U/\partial V)_T$

From property 2,

$$\left(\frac{\partial U}{\partial V}\right)_T = \left(\frac{\partial U}{\partial V}\right)_S + \left(\frac{\partial U}{\partial S}\right)_V \left(\frac{\partial S}{\partial V}\right)_T$$

Since $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$,

$$\pi_T = -p + T\left(\frac{\partial S}{\partial V}\right)_T$$

Topic 3E

Thermodynamic Eq. of State

$$\pi_T = T\left(\frac{\partial S}{\partial V}\right)_T - p$$

From the Maxwell relation $(\partial S/\partial V)_T = (\partial p/\partial T)_V$,

$$\pi_T = T\left(\frac{\partial p}{\partial T}\right)_V - p$$

or,

$$p = T\left(\frac{\partial p}{\partial T}\right)_V - \pi_T$$

(thermodynamic equation of state)

Example 3E.2

Show thermodynamically that $\pi_T = 0$ for a perfect gas, and compute its value for a van der Waals gas.

For a perfect gas,

$$\left(\frac{\partial p}{\partial T}\right)_V = \left[\frac{\partial}{\partial T}\left(\frac{nRT}{V}\right)\right]_V = \frac{nR}{V}$$

From the thermodynamic equation of state,

$$\pi_T = T\left(\frac{\partial p}{\partial T}\right)_V - p = T\left(\frac{nR}{V}\right) - p = \frac{nRT}{V} - p = p - p = 0$$

Example 3E.2

For a van der Waals gas,

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

$$\left(\frac{\partial p}{\partial T}\right)_V = \left[\frac{\partial}{\partial T}\left(\frac{nRT}{V - nb} - a\frac{n^2}{V^2}\right)\right]_V = \frac{nR}{V - nb}$$

From the thermodynamic equation of state,

$$\pi_T = T\left(\frac{\partial p}{\partial T}\right)_V - p = T\left(\frac{nR}{V - nb}\right) - p = \frac{nRT}{V - nb} - \left(\frac{nRT}{V - nb} - a\frac{n^2}{V^2}\right)$$

$$\pi_T = a\frac{n^2}{V^2}$$

Thermochemistry of G

$$G = H - TS$$

At constant T,

$$\Delta G = \Delta H - T\Delta S$$

We can define the standard Gibbs energy of reaction:

$$\Delta_r G^{\ominus} = \Delta_r H^{\ominus} - T\Delta_r S^{\ominus}$$

and the standard Gibbs energy of formation:

$$\Delta_f G^{\ominus} = \Delta_f H^{\ominus} - T\Delta_f S^{\ominus}$$

Standard Reaction Potentials

$$\Delta_r H^\ominus = \sum_{\text{products } i} \nu(i)\Delta_f H^\ominus(i) - \sum_{\text{reactants } j} \nu(j)\Delta_f H^\ominus(j)$$

$$\Delta_r S^\ominus = \sum_{\text{products } i} \nu(i) S_m^\ominus(i) - \sum_{\text{reactants } j} \nu(j) S_m^\ominus(j)$$

$$\Delta_r G^\ominus = \sum_{\text{products } i} \nu(i)\Delta_f G^\ominus(i) - \sum_{\text{reactants } j} \nu(j)\Delta_f G^\ominus(j)$$

Convention Again

$$\Delta_f G^{\ominus}(\mathrm{H}^+, aq) = 0$$

e.g. (Brief Illustration 3D.2)

$$(1/2)\mathrm{H_2}(g) + (1/2)\mathrm{Cl_2}(g) \rightarrow \mathrm{H^+}(aq) + \mathrm{Cl^-}(aq)$$

$$\Delta_r G^{\ominus} = \Delta_f G^{\ominus}(\mathrm{H}^+, aq) + \Delta_f G^{\ominus}(\mathrm{Cl}^-, aq) = \Delta_f G^{\ominus}(\mathrm{Cl}^-, aq)$$

Pressure Dependence of G

$$\mathrm{d}G = V\mathrm{d}p - S\mathrm{d}T$$

At constant T,

$$\mathrm{d}G = V\mathrm{d}p$$

Integration leads to

$$\int_{\text{state } i}^{\text{state } f} \mathrm{d}G = \int_{\text{state } i}^{\text{state } f} V\,\mathrm{d}p$$

$$\Delta G = G_f - G_i = \int_{p_i}^{p_f} V\,\mathrm{d}p$$

Pressure Dependence of G

$$\Delta G = \int_{p_i}^{p_f} V \, \mathrm{d}p$$

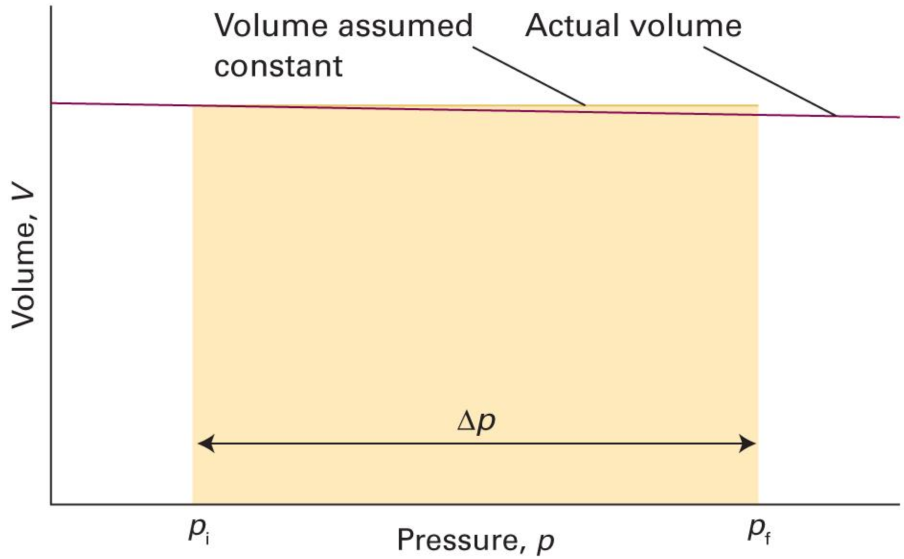
Dividing both sides by n,

$$\Delta G_\mathrm{m} = \int_{p_i}^{p_f} V_\mathrm{m} \, \mathrm{d}p$$

1. Condensed phase: $V_\mathrm{m} \approx$ constant
2. Gas phase: $V_\mathrm{m} = V/n = RT/p$

Condensed Phase

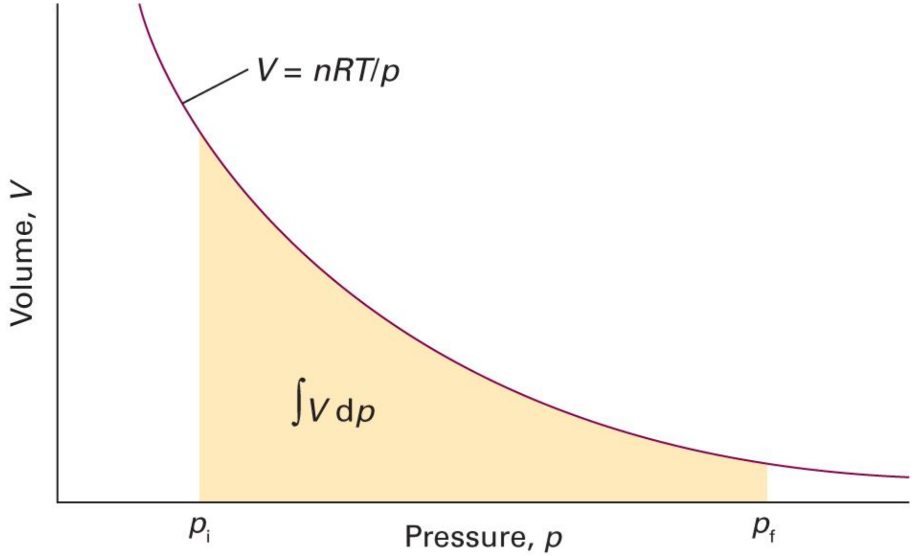
$$\Delta G_{\mathrm{m}} = \int_{p_i}^{p_f} V_{\mathrm{m}}\,\mathrm{d}p = V_{\mathrm{m}} \int_{p_i}^{p_f} \mathrm{d}p = V_{\mathrm{m}}(p_f - p_i)$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 3E.4)

Gas Phase

$$\Delta G_\mathrm{m} = \int_{p_i}^{p_f} V_\mathrm{m}\,\mathrm{d}p = \int_{p_i}^{p_f} \frac{RT}{p}\,\mathrm{d}p = RT\left(\ln p_f - \ln p_i\right)$$

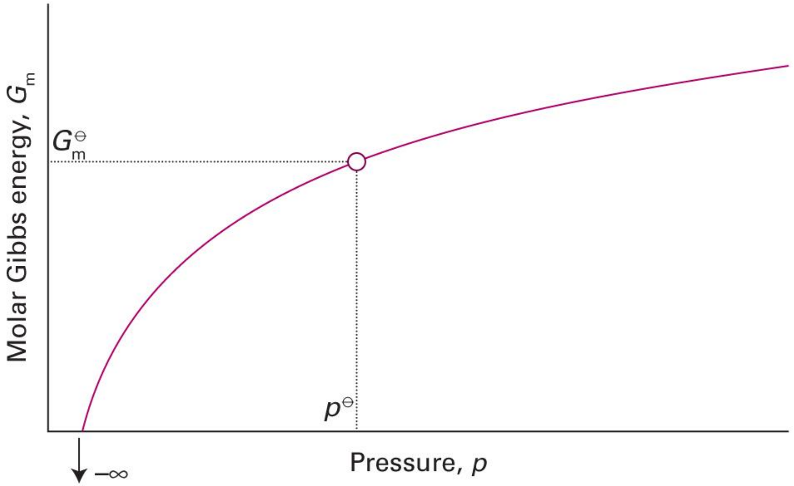


(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 3E.5)

Gas Phase: Standard State

$$\Delta G_\mathrm{m} = G_\mathrm{m}(p_f) - G_\mathrm{m}(p_i) = RT\ln\frac{p_f}{p_i}$$

$$G_\mathrm{m}(p) = G_\mathrm{m}^\ominus + RT\ln\frac{p}{p^\ominus}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 3E.6)

Summary

Thermodynamic potentials

Internal energy

U

Helmholtz energy

$A = U - TS$

Enthalpy

$H = U + pV$

Gibbs energy

$G = H - TS$

Clausius ineq. → **Spontaneity**

$\Delta A_{T,V} \leq 0$

$\Delta G_{T,p} \leq 0$

First law → **Maximum work**

$A_T = w_{\max}$

$G_{T,p} = w_{\text{add,max}}$

Differential forms

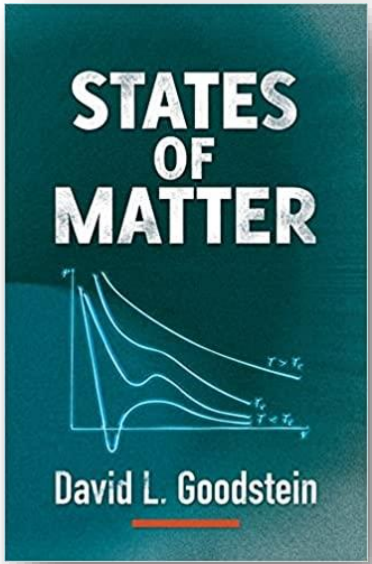
Applications

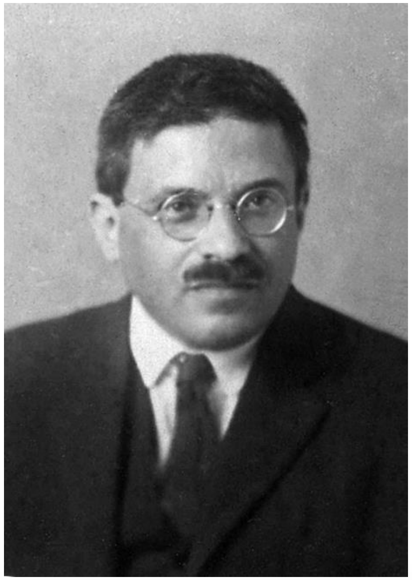
- Gibbs-Helmholtz equation
- Maxwell relations
- Thermodynamic equation of state

Thermochemistry

Pressure dependence

6. Statistical Mechanics





1.1 INTRODUCTION: THERMODYNAMICS AND STATISTICAL MECHANICS OF THE PERFECT GAS

Ludwig Boltzmann, who spent much of his life studying statistical mechanics, died in 1906, by his own hand. Paul Ehrenfest, carrying on the work, died similarly in 1933. Now it is our turn to study statistical mechanics.

Goodstein, *States of Matter* (1985)

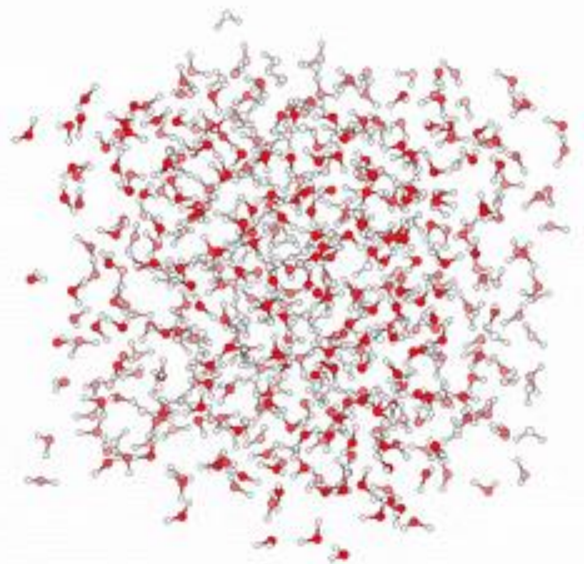
Chapter Overview

- Maxwell-Boltzmann statistics
  - Boltzmann factor
  - Partition function and thermodynamic quantities
- Classical statistical mechanics
  - Phase space
  - Calculation of partition function
- Applications
  - Perfect gas and real gas
  - Equipartition theorem

Not in Atkins

Statistical Mechanics

Thermodynamics (p, V, T, …)



"Statistics" of individual particles

(Image: https://commons.wikimedia.org/wiki/File:MD_water.gif)

Not in Atkins

Maxwell-Boltzmann Statistics

Average distribution of independent particles

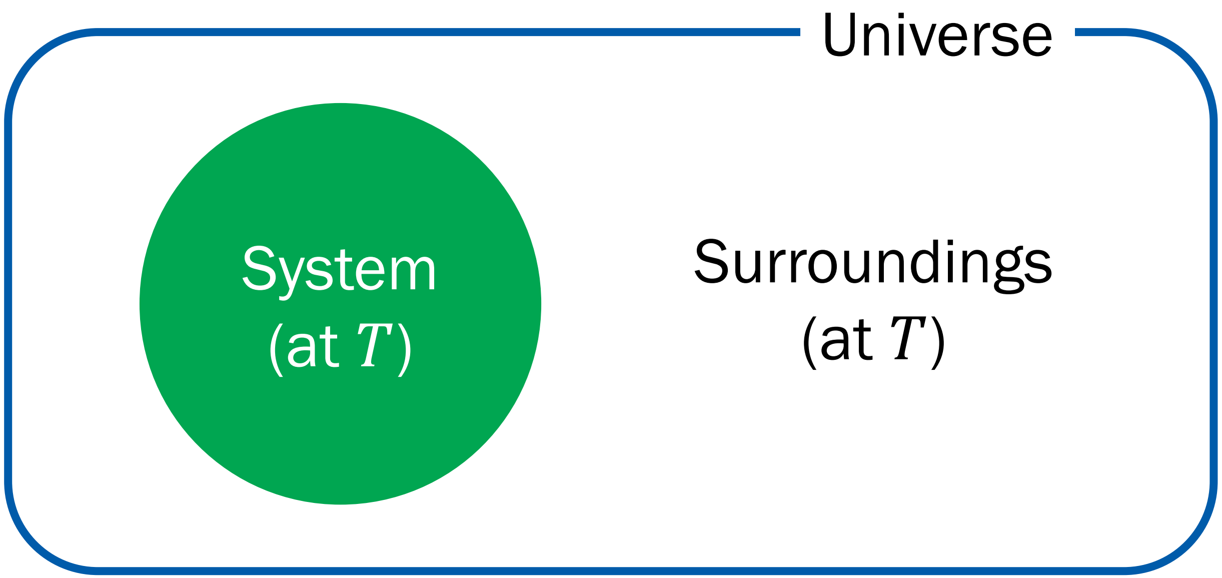
of a system in thermal equilibrium

Described by the Boltzmann factor

Not in Atkins

System Microstates

For a closed system with constant V and n,



Consider two microstates s_1 and s_2.

Not in Atkins

Probabilities

- Probability of having s_i is proportional to the number of microstates in the surroundings

$$P(s_i) \propto W_{\mathrm{surr}}(s_i)$$

- Ratio between probabilities of having s_1 and s_2

$$\frac{P(s_2)}{P(s_1)} = \frac{W_{\mathrm{surr}}(s_2)}{W_{\mathrm{surr}}(s_1)}$$

Not in Atkins

Entropy Comes in

$$\frac{P(s_2)}{P(s_1)} = \frac{W_{\mathrm{surr}}(s_2)}{W_{\mathrm{surr}}(s_1)}$$

Using the statistical definition of entropy, $S = k_{\mathrm{B}} \ln W$,

$$\frac{P(s_2)}{P(s_1)} = \frac{e^{S_{\mathrm{surr}}(s_2)/k_{\mathrm{B}}}}{e^{S_{\mathrm{surr}}(s_1)/k_{\mathrm{B}}}} = e^{\{S_{\mathrm{surr}}(s_2) - S_{\mathrm{surr}}(s_1)\}/k_{\mathrm{B}}}$$

What is $S_{\mathrm{surr}}(s_2) - S_{\mathrm{surr}}(s_1)$?

Not in Atkins

Entropy of Surroundings

Recall the fundamental equation of thermodynamics:

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V = T\mathrm{d}S \text{ (at constant } V\text{)}$$

For constant T, $\Delta U = T\Delta S$ and

$$S_{\mathrm{surr}}(s_2) - S_{\mathrm{surr}}(s_1) = \frac{1}{T}\{U_{\mathrm{surr}}(s_2) - U_{\mathrm{surr}}(s_1)\}$$

$$= -\frac{1}{T}\{U(s_2) - U(s_1)\}$$

Not in Atkins

Written in System Variables

$$\frac{P(s_2)}{P(s_1)} = e^{\{S_{\mathrm{surr}}(s_2) - S_{\mathrm{surr}}(s_1)\}/k_{\mathrm{B}}}$$

$$S_{\mathrm{surr}}(s_2) - S_{\mathrm{surr}}(s_1) = -\frac{1}{T}\{U(s_2) - U(s_1)\}$$

Combining the two equations,

$$\frac{P(s_2)}{P(s_1)} = e^{-\{U(s_2) - U(s_1)\}/k_{\mathrm{B}}T} = \frac{e^{-U(s_2)/k_{\mathrm{B}}T}}{e^{-U(s_1)/k_{\mathrm{B}}T}}$$

Topic 12A

Boltzmann Factor

$$\frac{P(s_2)}{P(s_1)} = \frac{e^{-U(s_2)/k_\mathrm{B}T}}{e^{-U(s_1)/k_\mathrm{B}T}}$$

In general, for a microstate s with energy E,

$$P(s) \propto e^{-E(s)/k_\mathrm{B}T}$$ (Boltzmann factor)